Campus Vehicle Pass System

Category B Practice - Part A, Problem 2

Problem summary

Yearly fees: Bike 300, Car 1000, E-Bike 300, E-Car 1000. Only electric vehicles can book charging.

Vehicle subclasses provide type/fee; the Chargeable interface models the capability.

Java solution

```
import java.util.*;
public class Main {
 interface Chargeable {boolean bookChargingBay();}
 abstract static class Vehicle {final String pass,owner;Vehicle(String p,String o){pass=p;owner=o;}abstract Stri
 static class Bike extends Vehicle {Bike(String p,String o){super(p,o);}String type(){return "Bike";}int fee(){r
 static class Car extends Vehicle {Car(String p,String o){super(p,o);}String type(){return "Car";}int fee(){retu
 static class EBike extends Vehicle implements Chargeable {EBike(String p,String o){super(p,o);}String type(){re
 static class ECar extends Vehicle implements Chargeable {ECar(String p,String o){super(p,o);}String type(){retu
 static Vehicle make(String t,String p,String o){switch(t.toLowerCase()){case "bike":return new Bike(p,o);case "
 public static void main(String[] args){Scanner sc=new Scanner(System.in);Map<String,Vehicle> m=new HashMap<>();
  while(sc.hasNextLine()){String line=sc.nextLine().trim();if(line.isEmpty())continue;String[] p=line.split("\\s
   if(p[0].equals("PASS")&&p.length>=4){String owner=String.join(" ",Arrays.copyOfRange(p,3,p.length));Vehicle v
   else if(p[0].equals("CHARGE")&&p.length==2){Vehicle v=m.get(p[1]);if(v==null)System.out.println("Vehicle not
  }
 }
}
```

Example input

```
PASS Bike KA01 Asha
PASS ECar KA02 Ravi
CHARGE KA02
CHARGE KA01
```

Expected output

```
KA01 (Bike) pass fee 300
KA02 (ECar) pass fee 1000
KA02 charging bay allotted
KA01 rejected: charging unsupported
```

Notes

An interface expresses a CAN-DO capability; non-electric vehicle classes do not implement Chargeable.